

Multilabel Coronary Heart Disease Screening Using an Expert-Designed Instrument and Multimodal Knowledge Enhancement

Jia Li¹ and Feng Gao¹

School of Computer Science and Technology, Wuhan University of Science and Technology,
Wuhan, China
18671640800@163.com

Abstract. Multiple conditions related to coronary heart disease (CHD) may coexist in patients undergoing clinical evaluation, creating a need for low-cost preliminary multilabel assessment before confirmatory testing. Existing coronary prediction studies commonly focus on a single outcome or depend on electrocardiography, laboratory testing, advanced imaging, or longitudinal electronic health records; their applicability to a low-cost, single-encounter multilabel setting therefore remains uncertain. This study evaluated a non-invasive, knowledge-enhanced framework for predicting eight CHD-related labels in a retrospective cohort of 367 patients from one tertiary hospital. The inputs comprised an expert-designed 52-item structured clinical assessment and ten geometric features derived from bilateral auricular photographs. The framework extends patient-conditioned graph attention by combining a background medical knowledge graph with expert-curated feature-to-disease relations, while class weighting, disease-aware bidirectional hard-example sampling, and validation-selected disease-specific thresholds address label imbalance. In descriptive cross-validation comparisons, the framework achieved the highest point estimates for macro-area under the precision-recall curve (0.565), macro-F1 (0.543), and macro-recall (0.741) among the evaluated supervised methods, whereas BR-TabPFNv2 achieved higher macro-AUROC, macro-precision, and macro-specificity. The macro-specificity of 0.513 indicates that fewer missed positive labels were accompanied by more false-positive predictions. These findings provide preliminary evidence for low-cost knowledge-enhanced assessment in the specified clinical cohort rather than uniform performance superiority, and require prospective multicenter validation before clinical use.

Keywords: coronary heart disease · multilabel classification · structured clinical assessment · multimodal learning · medical knowledge graph · class imbalance

1 Introduction

Coronary heart disease (CHD) comprises a heterogeneous spectrum that includes chronic and acute manifestations. Unstable angina, non-ST-segment elevation myocardial infarction, ST-segment elevation myocardial infarction, stable angina, ischemic cardiomyopathy, and other CHD-related conditions may coexist in one patient. Current

guidelines therefore recommend stepwise assessment using symptoms, medical history, physical examination, electrocardiographic findings, biomarkers, and anatomical or functional testing [36, 6]. Artificial intelligence is increasingly studied across this pathway, but its clinical role depends on the population being assessed and on the information available at the time of prediction [5]. The present study concerns patients already undergoing hospital evaluation for suspected CHD and addresses preliminary assessment before confirmatory evaluation rather than population-wide screening or replacement of standard diagnosis.

Low-cost prediction from a single clinical encounter poses a distinct problem. A fixed set of questions can capture demographic characteristics, symptoms, medical history, risk factors, lifestyle, and selected findings that are available early in the clinical pathway. In contrast, many contemporary cardiovascular prediction systems rely on electrocardiographic signals, laboratory variables, advanced imaging, or longitudinal electronic health records [39, 32, 34, 40]. These sources can be informative but are not always available when an initial assessment is made. Moreover, much of the cited work targets one diagnosis or one subsequent clinical event rather than several potentially coexisting CHD-related labels. Evidence for low-cost, single-encounter multilabel prediction consequently remains limited.

Auricular morphology illustrates both the opportunity and the limitations of an accessible external sign. Earlobe creases have been associated with coronary atherosclerosis in angiography-based cohorts [29]. Observational evidence also relates diagonal earlobe creases to age, sex, obesity, and hypertension, indicating substantial potential for confounding [26]. A systematic review of diagnostic accuracy concluded that the presence or absence of a crease changes pre-test probability only modestly and should not determine patient management by itself [37]. The relevant question is therefore not whether auricular morphology can independently diagnose CHD, but whether quantitative, interpretable auricular measurements provide incremental information when combined with structured clinical variables.

Knowledge-guided learning offers one way to integrate heterogeneous but individually weak predictors in a small cohort. PLATO aligns tabular features with an auxiliary knowledge graph and uses graph propagation to generate patient-conditioned first-layer weights [28]. Other systems construct patient-specific graphs, combine ontologies with clinical text, retrieve external knowledge, or enrich medical concepts before prediction [18, 38, 24, 20]. These approaches demonstrate several mechanisms for incorporating medical knowledge, but they mainly operate on longitudinal records, medical codes, or retrieved text. They do not directly establish how a background graph and task-specific, expert-confirmed feature-to-disease relations should be combined for one structured suspected-CHD encounter.

Multilabel prediction introduces a separate learning problem because disease prevalences and error profiles differ substantially. A common training distribution and a fixed threshold of 0.5 need not be appropriate for every label. Prior work has used external knowledge, class-aware augmentation, hard-sample selection, and label-specific thresholding to improve learning for infrequent labels [11, 31, 41, 22]. These techniques motivate the class weighting, disease-aware bidirectional sampling, and validation-selected

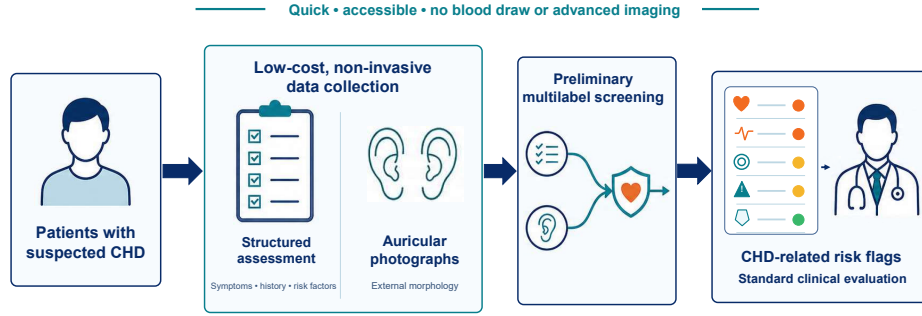


Fig. 1. Low-cost, non-invasive preliminary screening pathway for patients with suspected CHD.

thresholds used here. They are treated as supporting optimization components rather than as the principal source of methodological novelty.

Taken together, the literature leaves three related questions insufficiently resolved in the present clinical setting. First, it is unclear whether information obtainable from one low-cost encounter can support prediction of several coexisting CHD-related conditions. Second, the incremental value of interpretable auricular measurements beyond a fixed structured clinical assessment remains uncertain. Third, existing knowledge-guided models do not determine whether expert-confirmed relations between clinical features and disease nodes improve a PLATO-based representation in a small, imbalanced cohort. This study addresses these questions in 367 patients undergoing evaluation for suspected CHD at a single tertiary hospital.

The study makes three scoped contributions. First, it defines a standardized data-acquisition and representation scheme that combines an expert-designed 52-item clinical assessment with ten normalized geometric variables derived from bilateral auricular photographs. The instrument has no aggregate score; its responses remain individual model predictors. Second, it extends the background-graph representation with eight disease nodes and an expert-curated clinical diagnostic knowledge layer in which typed relations connect questionnaire-derived features to prediction targets. Auricular variables are aligned with the background medical entity space, and the two acquisition channels are fused after interpretable feature extraction. Third, it evaluates this task-specific integration against conventional, recent tabular, knowledge-guided, and direct large-language-model comparators, and separates the effects of auricular features, clinical diagnostic knowledge, imbalance-aware training, and disease-specific thresholds through controlled ablations. The evaluation is intended to characterize feasibility and the recall–specificity trade-off, not to claim uniform superiority or immediate clinical deployability.

Figure 1 summarizes the proposed pathway. The predictor set consists only of structured responses and auricular measurements. Laboratory biomarkers, coronary computed tomography angiography (CCTA), and invasive coronary angiography may contribute to the clinical reference diagnosis but remain outside the model input.

2 Related Work

2.1 Low-Cost Coronary Screening with Structured and Physical-Sign Data

Clinical evaluation of suspected CHD begins with structured reasoning. Contemporary guidance places symptoms, medical history, physical examination, resting electrocardiography, and basic testing before more advanced investigations when indicated [36, 6]. Machine-learning studies have attempted to reduce the information burden at this stage. For example, a model based on nine readily available clinical variables predicted severe coronary calcification in patients referred for angiography [17]. In a related resource-conscious setting, a cost analysis found that staged use of AI-enabled electrocardiography and handheld ultrasound could reduce the immediate cost per detected case of structural heart disease [30]. These studies show the practical value of accessible inputs, but their targets and acquisition pathways differ from simultaneous prediction of several CHD-related conditions using a fixed clinical assessment and an external physical sign.

A structured instrument can make low-cost data collection reproducible, although its intended function must be specified. COSMIN guidance defines patient-reported outcome measures in terms of measurement properties and interpretable scores [25]. The 52-item instrument used in this study instead records expert-selected observations as separate predictors and has no total score, latent construct, or diagnostic cutoff. It should therefore be understood as a standardized clinical data-acquisition instrument rather than as a validated psychometric or stand-alone diagnostic scale.

External physical signs provide inexpensive adjunctive information but require cautious interpretation. Earlobe creases have been associated with the presence, extent, and severity of angiographic coronary atherosclerosis [29]. Their reported diagnostic accuracy nevertheless varies, and a systematic review found insufficient evidence for independent clinical decision-making [37]. Associations with age, sex, obesity, and hypertension further indicate that the crease may partly reflect conventional cardiovascular risk factors [26]. The present study consequently represents bilateral auricular morphology through quantitative measurements and tests its incremental value alongside clinical variables rather than treating it as an independent diagnostic marker. This distinction separates the proposed use of auricular information from studies that evaluate a visible crease as a binary sign.

2.2 Knowledge-Guided Learning from Multiple Clinical Sources

Multimodal cardiovascular prediction combines sources with different acquisition processes. A recent review identified the fusion of clinical records, physiological signals, and imaging as a major direction in cardiovascular artificial intelligence [39]. Representative systems combine electronic health records with electrocardiograms for post-intervention outcome prediction [32], use electrocardiograms to predict several adverse cardiovascular events [23], or fuse clinical variables with electrocardiographic data for CHD prediction [40]. Tu et al. integrated CCTA, clinical variables, and electrocardiograms while accounting for missing modalities [34]. These systems demonstrate the potential of multimodal fusion, but they generally depend on raw physiological signals, advanced

imaging, or longitudinal records. The present study addresses a narrower setting in which two acquisition channels are converted into structured variables from one encounter: fixed-option clinical responses and ten interpretable auricular measurements.

Representation determines whether such heterogeneous variables can share a model. AutoFM searches modality-specific encoders and fusion structures for electronic health record inputs [10]. The present study does not perform end-to-end image learning; instead, the photographic channel is converted into numeric morphology before feature-level fusion. Acquisition is therefore multimodal, whereas the model input is a unified tabular representation. This design preserves the origin and interpretation of the auricular variables and allows each retained feature to be aligned with a medical entity.

Knowledge graphs offer a complementary mechanism for structuring heterogeneous predictors. Their construction requires source selection, entity and relation extraction, normalization, fusion, and quality assessment [1]. Large language models can assist candidate extraction, but biomedical evaluations report variable reliability and hallucinated content, making provenance control and expert review important when graph structure influences clinical prediction [3]. This distinction is relevant here because automatically extracted candidates are not accepted directly into the task graph.

Existing knowledge-enhanced predictors can be grouped by how knowledge enters the model. PLATO aligns tabular features with graph nodes and generates patient-conditioned feature weights through message passing [28]. Patient- and record-oriented methods instead build personalized or heterogeneous EHR graphs, as in GraphCare and MHGRL [18, 24]. Retrieval-oriented systems, including RAM-EHR and KARE, select concept-relevant knowledge or graph communities to support prediction and reasoning [38, 19]. These approaches primarily operate on longitudinal records, medical codes, clinical notes, or retrieved text rather than a fixed set of variables collected during one suspected-CHD encounter.

Graph-based clinical prediction has also become more closely connected with language-model reasoning. medIKAL uses graph search to refine diagnoses generated from electronic medical records [16]; MedCo enriches medical concept graphs with statistically supported and language-model-inferred relations and rationales [20]; and ClinPath represents longitudinal patient-provider interactions in a general-purpose clinical graph [2]. ReMedi generates reasoning examples for EHR outcome prediction without constructing an explicit expert diagnostic graph [7]. Generated or retrieved knowledge can broaden coverage, whereas a small diagnostic cohort also benefits from task-specific relations whose clinical meaning and provenance can be inspected. The present framework therefore retains PLATO’s background feature alignment but adds expert-confirmed relations between questionnaire-derived features and eight disease nodes. Auricular variables participate through alignment with the background graph rather than through additional disease links. This layered design differs from both generic EHR graphs and automatically generated diagnostic reasoning.

Knowledge graphs have also been studied in medical image analysis, where limited multimodal integration and weak interpretability remain recurring concerns [15]. Converting ear photographs into explicit measurements does not exploit the full information available to an image encoder, but it avoids treating an external sign as an opaque diagnostic image and permits direct component ablation. The resulting contribution is

therefore a task-specific, knowledge-aligned integration strategy rather than a general image-analysis architecture.

2.3 Imbalance-Aware Multilabel Learning as Supporting Optimization

Multilabel clinical data are imbalanced both across diseases and between positive and negative examples within each disease. DKEC indicates that external medical knowledge can be particularly useful for few-shot diagnosis labels [11]. Other approaches use class-aware data mixing or label correction [31], while adaptive batch selection emphasizes hard examples associated with minority labels [41]. The disease-aware sampling procedure in this study belongs to this broader optimization family. It changes the frequency with which observed training records are presented but does not synthesize new patients or introduce external labels.

Decision thresholds address a different part of the prediction pipeline. A common threshold can be suboptimal for infrequent labels, but unconstrained direct F-score maximization on a small validation set can overfit [22]. Disease-specific thresholds should therefore be selected using validation data and fixed before test evaluation. This distinction also determines how results are interpreted: threshold selection can change precision, recall, specificity, and F1 at an operating point, but it cannot change AUROC or AUPRC when the underlying prediction scores remain fixed.

Comparisons with conventional and recent tabular learners are necessary because no single architecture dominates structured datasets. Carefully configured multilayer perceptrons and boosted trees remain competitive [14]. TabM uses parameter-efficient ensembling for tabular deep learning [12], whereas TabPFN uses prior-data pretraining for small to medium tabular datasets [13]. These methods provide distinct references for assessing whether the knowledge-aligned representation adds value beyond general-purpose tabular learning. PLATO provides the more direct architectural reference because it uses the same underlying principle of aligning tabular variables with an auxiliary knowledge graph [28].

The present study differs from these research lines in four respects. It targets eight potentially coexisting labels in patients already undergoing evaluation for suspected CHD; uses one fixed clinical encounter rather than longitudinal records; represents the photographic channel through interpretable geometry rather than a raw image encoder; and combines background entity alignment with an expert-confirmed clinical feature-to-disease topology. Class weighting, hard-example sampling, and disease-specific thresholds support this representation but are not claimed as independently novel. The evaluation accordingly separates image-derived features, clinical diagnostic knowledge, training reweighting, sampling, and threshold selection through controlled ablations. It also reports ranking metrics and threshold-dependent metrics separately so that improved positive-label detection is not conflated with uniform superiority. This scope is consistent with recent recommendations for transparent reporting, applicability assessment, and deployment-aware interpretation of clinical prediction models [9, 27, 21].

3 Method

3.1 Study Design and Multilabel Problem Formulation

The retrospective cohort was assembled from the cardiovascular ward of one tertiary Grade A hospital. Four hundred patients undergoing clinical evaluation for suspected CHD were screened. After 33 exclusions, 367 remained. Eligibility required a clinical suspicion of CHD based on presenting symptoms, medical history, or preliminary clinical findings. Records with more than 70% missing features and duplicate hospitalizations were excluded. For a patient with repeated visits, only the first record was retained.

The eight labels were coronary atherosclerosis (AS), coronary atherosclerotic heart disease (CHD), unstable angina (UA), ST-segment elevation myocardial infarction (STEMI), non-ST-segment elevation myocardial infarction (NSTEMI), stable angina (SA), silent or asymptomatic myocardial ischemia (SMI), and ischemic cardiomyopathy (ICM). Labels could coexist. The reference labels were assigned by medical professors on the basis of their clinical judgment.

Let patient i have questionnaire variables $x_i^{(q)} \in \mathbb{R}^{D_q}$, auricular image-derived variables $x_i^{(a)} \in \mathbb{R}^{D_a}$, and a binary target $y_i \in \{0, 1\}^K$. Before categorical expansion, $D_q = 52$, $D_a = 10$, and $K = 8$. The raw predictors are

$$x_i^{(\text{raw})} = \begin{bmatrix} x_i^{(q)}; x_i^{(a)} \end{bmatrix}. \quad (1)$$

After encoding, normalization, and entity alignment, the retained vector is $x_i \in \mathbb{R}^D$ with $D = 168$. For disease k , the network produces $p_{ik} = \sigma(z_{ik})$, and a disease-specific threshold τ_k yields

$$\hat{y}_{ik} = \mathbb{I}(p_{ik} \geq \tau_k). \quad (2)$$

The output is intended for preliminary eight-label screening before confirmatory clinical evaluation. Only questionnaire responses and auricular measurements enter the predictor. Examination reports and advanced tests may contribute to the reference labels but are not predictor inputs.

Figure 2 summarizes the complete prediction and optimization pipeline.

3.2 Expert-Designed Screening Instrument and Multimodal Input Representation

Medical professors and clinical experts designed the 52-item instrument using medical guideline knowledge and clinical experience. The items were discussed and revised before data collection, and every question has predefined response options. The content covers several categories, including personal information, symptoms of chest pain or chest tightness, accompanying symptoms, medical history, and lifestyle habits. Electrocardiographic findings are recorded as one structured item within the instrument.

The instrument is unscored. Each response remains an independent predictor after encoding; no total score or diagnostic cutoff is calculated. It therefore functions as a

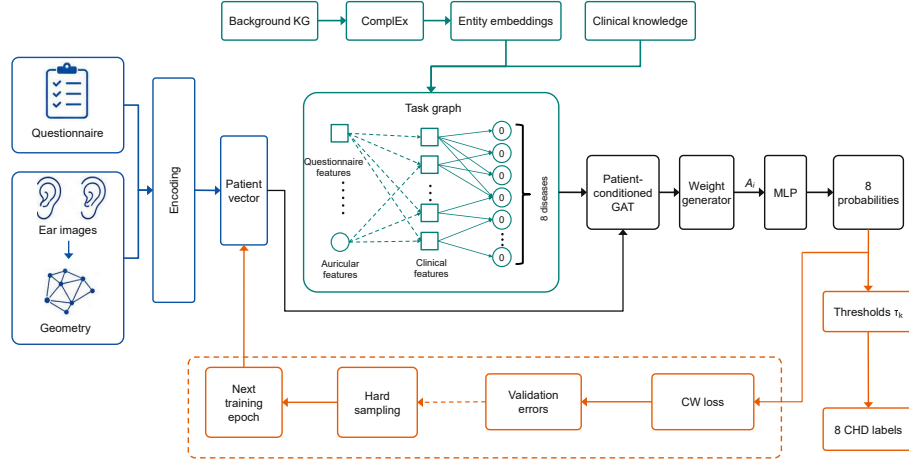


Fig. 2. Architecture of the multimodal knowledge-enhanced PLATO framework. Questionnaire variables and auricular geometry are encoded into the patient vector; background-graph embeddings and expert-curated questionnaire-to-disease links form the task graph. The patient-conditioned graph-attention network generates first-layer weights for the eight-label predictor, while class-weighted loss, disease-aware hard-example sampling, and validation-based disease-specific thresholds address label imbalance.

structured clinical screening and data-collection instrument rather than a psychometric scale that measures one latent construct.

Two additional photographs were available for each patient, one showing the complete left ear and the other the complete right ear. Image processing represented the auricular crease in each photograph by width, height, area, aspect ratio, and relative distance. The bilateral image channel thus contributed ten continuous variables. Width, height, and area were normalized, while aspect ratio and relative distance were included as derived quantities. Converting the photographs into compact geometric measurements makes the image information interpretable and permits direct alignment with medical entities.

Patient identifiers were used only to align feature and label tables and were excluded from the model. The 52 questionnaire variables and ten auricular variables formed 62 raw predictors. Categorical fields were one-hot encoded, and three multi-select questions were split before binary expansion. Continuous variables were normalized. Nonparseable continuous values and residual missing values were assigned zero according to the preprocessing rule of the implementation.

Each encoded feature name was matched exactly to a medical entity. Unmatched features were filtered before training. The resulting 168 columns, including the encoded auricular measurements, all had corresponding entities in the background knowledge graph, producing $X \in \mathbb{R}^{367 \times 168}$.

3.3 Dual-Layer Medical Knowledge Construction

Background medical knowledge. The auxiliary graph $G = (V, R, E)$ was constructed from cardiovascular guidelines and medical textbooks. A large language model extracted candidate entities, relation types, and triples, but did not determine the final graph content. Rule-based processing and manual normalization unified synonymous disease and feature names. Cardiovascular experts then reviewed the task-relevant knowledge. This separation between candidate extraction and expert verification addresses current concerns about the validity of biomedical knowledge graphs [1, 3].

CompLex learned embeddings for entities in the auxiliary graph [33]. The complex embedding dimension was 100, training lasted 100 epochs, and the random seed was 42. Concatenating the real and imaginary parts produced a 200-dimensional real vector for every entity. Because the embeddings served only as downstream node initialization, CompLex was trained on the complete auxiliary graph without a held-out link-prediction split. Relation embeddings remained confined to this stage. Let $V_x \in \mathbb{R}^{D \times d_e}$ contain the embeddings of the $D = 168$ aligned feature entities, where $d_e = 200$, and let E_x denote their induced background subgraph.

Expert-curated clinical diagnostic knowledge. A second knowledge layer represents how questionnaire-derived clinical features relate to the eight prediction targets. Its candidate feature-to-disease connections were derived from guideline-based diagnostic knowledge and clinical experience, then discussed, revised, and confirmed by cardiovascular experts. The content covers evidence relations, diagnostic criteria, risk factors, symptom triggers, relief factors, subtype links, and disease progression.

The curated connections are stored as typed knowledge triples, with relation labels documenting their clinical meaning. The downstream graph-attention network uses the resulting topology for message passing and shares attention parameters across relation categories. Clinical rules consequently enter the model as expert-defined graph connections and constitute one form of medical knowledge. For example, smoking is connected to CHD as a risk factor, whereas rapid symptom relief after nitroglycerin is connected to stable angina as a relief factor.

The task graph contains the aligned feature entities and eight disease entities. Direct disease connections are defined for questionnaire-derived clinical features. Auricular image-derived features participate through entity alignment with the background graph and do not receive additional feature-to-disease connections in this layer. Duplicate connections are removed, every directed connection receives a reverse counterpart, and self-loops are added by the graph-attention layer.

Disease nodes provide semantic embeddings while their patient observation entries are set to zero. For patient i , the augmented input and embedding matrices are

$$x'_i = [x_i; \mathbf{0}_K], \quad V'_x = [V_x; V_d], \quad (3)$$

where V_d contains the eight disease embeddings. This construction allows disease semantics to influence connected feature representations without inserting unknown labels into the patient vector.

3.4 Multimodal Knowledge-Enhanced PLATO

The prediction backbone follows PLATO [28]. A graph-attention network propagates fixed entity embeddings over the induced feature graph and uses the patient vector to condition node representations [35]. A trainable mapping g_θ converts these representations into a patient-specific first-layer weight matrix:

$$\begin{aligned}\tilde{V}_i &= \text{GAT}(x_i, E_x, V_x), \\ A_i &= g_\theta(\tilde{V}_i) \in \mathbb{R}^{D \times H_1}, \\ h_i^{(1)} &= x_i^\top A_i / s.\end{aligned}\tag{4}$$

ReLU-activated linear layers map $h_i^{(1)}$ to eight logits. After clinical diagnostic knowledge is incorporated, the corresponding computation becomes

$$\begin{aligned}\tilde{V}'_i &= \text{GAT}(x'_i, E'_x, V'_x), \\ A'_i &= g_\theta(\tilde{V}'_i), \\ h_i^{(1)} &= (x'_i)^\top A'_i / s,\end{aligned}\tag{5}$$

where E'_x combines the background subgraph with the expert-curated clinical diagnostic topology, reverse connections, and self-loops. Zero-valued disease nodes affect prediction indirectly. Their embeddings alter the representations of adjacent clinical features and, in turn, the generated first-layer weights.

In this study, multimodal knowledge enhancement comprises two complementary components. Auricular image-derived feature enhancement supplies a second acquisition channel and aligns its ten measurements with background-graph entities. Clinical diagnostic knowledge enhancement introduces disease nodes and expert-curated connections for questionnaire-derived features. The two components are fused within the patient-conditioned PLATO representation. Here, *multimodal* refers to questionnaire and photographic acquisition, with image fusion occurring after interpretable feature extraction.

3.5 Training and Disease-Specific Decision Optimization

Imbalance-aware training. Positive class weighting and disease-aware bidirectional hard-example sampling support learning from unequal label distributions. They only reweight or resample observed records from the current training subset. For disease k , the positive weight is

$$w_k^+ = \frac{N_k^-}{N_k^+ + \varepsilon},\tag{6}$$

where N_k^+ and N_k^- are calculated from that subset. If it contains no positive case, the weight returns to its neutral value. The weighted binary cross-entropy is

$$\mathcal{L}_{\text{WBCE}} = -\frac{1}{N_{\text{tr}}K} \sum_{i \in I_{\text{tr}}} \sum_{k=1}^K [w_k^+ y_{ik} \log p_{ik} + (1 - y_{ik}) \log(1 - p_{ik})].\tag{7}$$

At the end of each epoch, recall, specificity, false-positive rate, false-negative rate, F1, and precision are computed on the internal validation subset at a provisional threshold of 0.5. These measurements describe whether a disease is dominated by missed positives, false alarms, errors in both directions, or a stable error profile. Direction-specific pressures are

$$\begin{aligned} q_k^+ &= [\text{FNR}_k - t_{\text{FNR},k}]_+ + [t_{R,k} - \text{Recall}_k]_+, \\ q_k^- &= [\text{FPR}_k - t_{\text{FPR},k}]_+ + [t_{S,k} - \text{Specificity}_k]_+. \end{aligned} \quad (8)$$

The pressures update bounded positive and negative sampling intensities, α_k^+ and α_k^- . Training predictions then locate hard or boundary examples in both classes. If b_{ik}^+ and b_{ik}^- denote the corresponding state-dependent gains, the patient weight is

$$u_i = 1 + \text{clip}\left(\max_k b_{ik}^+ + \max_k b_{ik}^-, 0, B_{\max}\right), \quad \tilde{u}_i = \frac{u_i}{N_{\text{tr}}^{-1} \sum_{j \in \mathcal{I}_{\text{tr}}} u_j}. \quad (9)$$

The next epoch samples the training subset with replacement according to $\tilde{u}_i$. Validation performance determines the error direction and updates the sampling intensities, while training predictions determine which observed records receive additional weight. Outer-test labels are not used.

Disease-specific decision thresholds. Threshold selection begins after restoration of the checkpoint with the best internal-validation score. Each disease is evaluated on $T = \{0.10, 0.15, \dots, 0.90\}$. For candidate t , the base utility is

$$U_k(t) = 0.40F1_k(t) + 0.25R_k(t) + 0.20S_k(t) + 0.15P_k(t), \quad (10)$$

where R , S , and P denote recall, specificity, and precision. Soft targets for recall and specificity depend on the error pattern observed during training. With targets (R_k^*, S_k^*) , the candidate score is

$$\begin{aligned} J_k(t) &= U_k(t) - \lambda_c ([R_k^* - R_k(t)]_+ + [S_k^* - S_k(t)]_+) \\ &\quad - \lambda_d |t - 0.5|. \end{aligned} \quad (11)$$

The preset configuration uses $\lambda_c = 2.0$ and $\lambda_d = 0.05$. The threshold maximizing $J_k(t)$ is selected; ties favor the smaller target violation. If the validation subset contains fewer than two positive or two negative examples for a disease, τ_k returns to 0.5. All thresholds are frozen before evaluation on the corresponding outer test fold.

4 Experiments

4.1 Dataset and Experimental Setup

Table 1 summarizes the positive-label distribution in the full cohort. Because the task is multilabel, one patient may contribute to more than one disease category and the percentages do not sum to 100%. The distribution was markedly imbalanced: CHD had only 12 negative patients, whereas STEMI and NSTEMI had only 27 and 33 positive

Table 1. Distribution of positive disease labels in the study cohort.

Abbreviation	Disease label	Positive patients, n (%)
AS	Coronary atherosclerosis	72 (19.6)
CHD	Coronary atherosclerotic heart disease	355 (96.7)
UA	Unstable angina	70 (19.1)
STEMI	ST-segment elevation myocardial infarction	27 (7.4)
NSTEMI	Non-ST-segment elevation myocardial infarction	33 (9.0)
SA	Stable angina	180 (49.0)
SMI	Silent or asymptomatic myocardial ischemia	87 (23.7)
ICM	Ischemic cardiomyopathy	237 (64.6)

patients, respectively. This imbalance motivated the use of macro-averaged metrics and imbalance-aware training.

The experiments used the retrospective cohort of 367 patients described in Section 3.1. Five outer folds were constructed by multilabel stratification. Within each outer training portion, the same procedure formed five inner folds, one of which served as the validation subset. Depending on the outer fold, the training, validation, and test subsets contained approximately 234 to 236, 58 to 59, and 73 to 74 patients, respectively. All supervised methods used the same splits and preprocessing. Neural and knowledge-guided models were evaluated with three initializations per outer fold.

The controlled PLATO configurations were trained with Adam for 30 epochs using a learning rate of 0.001, a batch size of 32, and layer dimensions of [32, 32, 8]. The final proposed configuration used a learning rate of 0.002, layer dimensions of [64, 32, 8], and penalty coefficients $\lambda_c = 3.0$ and $\lambda_d = 0.08$. Model checkpoints were selected on the corresponding inner validation subset. In the primary comparison in Section 4.2, every supervised method that produced continuous prediction scores used the same validation-based disease-specific threshold protocol. The selected thresholds were fixed before outer-test evaluation. The direct LLM baselines returned discrete labels and therefore did not undergo threshold optimization. The controlled ablation in Section 4.3 intentionally enabled or disabled DCT to isolate its incremental effect.

Macro-AUPRC was the primary ranking metric because the label distribution was imbalanced, with macro-AUROC reported as a complementary measure. Macro-precision, macro-recall, macro-specificity, and macro-F1 were calculated from the final binary predictions. The direct LLM baselines returned discrete labels, so AUPRC and AUROC were unavailable. Because the final proposed configuration was selected during exploratory development based on aggregate cross-validation performance and uncertainty estimates were not available for every comparator, the results are interpreted descriptively.

4.2 Comparison with Baseline Methods

Table 2 compares the proposed framework with four groups of methods that represent different modeling assumptions: conventional classifiers, neural tabular models, knowledge-guided learning, and direct LLM inference. Conventional classifiers test

whether the task can be addressed by relatively simple linear or tree-based decision functions. Neural tabular models represent recent approaches developed specifically for structured data, whereas PLATO provides the closest architectural baseline for assessing the complete knowledge-enhanced extension. Direct LLM inference is included as an additional reference for examining whether general or medical language-model knowledge can produce the predefined disease labels without cohort-specific supervised training.

Binary relevance (BR) decomposes the eight-label problem into eight independent binary classification tasks. BR-LR uses logistic regression as a linear baseline, whereas BR-RF and BR-XGBoost use bagged and boosted tree ensembles, respectively [4, 8]. The multilayer perceptron (MLP) provides a conventional neural baseline. These methods establish the performance attainable with widely used tabular classifiers before knowledge-guided modeling is introduced. TabM represents recent tabular deep learning through parameter-efficient ensembling [12], while BR-TabPFNv2 applies a pretrained tabular foundation model independently to the eight labels [13]. Together, these models provide stronger structured-data references than conventional classifiers alone.

PLATO is the most direct architectural baseline because it aligns tabular features with an auxiliary knowledge graph and generates patient-conditioned feature weights through graph propagation [28]. All supervised methods, including PLATO, received the same encoded questionnaire and auricular predictors. The proposed configuration retains the PLATO backbone but adds the expert-curated feature-to-disease topology and the imbalance-aware training and decision components. The PLATO comparison therefore evaluates the complete extension relative to the original knowledge-guided architecture, while the controlled experiments in Section 4.3 separate the contributions of individual components. All supervised methods were evaluated using the common validation-based disease-specific threshold protocol described in Section 4.1.

Baichuan-M3, GLM-5.2, and Qwen3.7-max form a separate direct-inference group. Baichuan-M3 is a medical LLM, whereas GLM-5.2 and Qwen3.7-max are general LLMs. They received structured patient information and returned subsets of the eight predefined disease labels without threshold optimization. Because these models were not fitted to the study cohort and produced discrete labels rather than continuous prediction scores, their results are interpreted as an operating-point comparison rather than as a directly equivalent supervised-learning comparison.

Among the supervised methods, the proposed framework achieved the highest macro-AUPRC of 0.565. It exceeded BR-TabPFNv2, the strongest alternative on this metric, by 0.034 and PLATO by 0.089. The proposed framework also produced the highest macro-F1 of 0.543 and macro-recall of 0.741. Relative to BR-TabPFNv2, these two measures were higher by 0.034 and 0.058, respectively. The differences relative to PLATO extended across all threshold-dependent metrics: macro-F1 was 0.081 higher, precision 0.129 higher, recall 0.117 higher, and specificity 0.206 higher. Within this descriptive analysis, the complete framework therefore showed greater positive-label detection, precision, and specificity than the original knowledge-graph backbone.

The results do not show uniform superiority. BR-TabPFNv2 obtained the highest macro-AUROC of 0.704, compared with 0.659 for the proposed framework, and its precision and specificity were also higher. TabM ranked between BR-XGBoost and BR-

Table 2. Overall comparison with supervised baseline methods and direct LLM inference. The best reported value in each metric is shown in bold.

Method	Macro-AUPRC	Macro-AUROC	Macro-F1	Macro-Precision	Macro-Recall	Macro-Specificity
<i>Conventional and tabular supervised models</i>						
BR-LR	0.428	0.602	0.417	0.381	0.526	0.646
BR-RF	0.451	0.621	0.421	0.409	0.533	0.661
BR-XGBoost	0.496	0.657	0.478	0.455	0.602	0.607
MLP	0.463	0.628	0.445	0.423	0.579	0.532
TabM	0.507	0.676	0.488	0.472	0.645	0.566
BR-TabPFNv2	0.531	0.704	0.509	0.526	0.683	0.594
<i>Direct LLM inference</i>						
Baichuan-M3	-	-	0.381	0.584	0.383	0.779
GLM-5.2	-	-	0.418	0.555	0.382	0.765
Qwen3.7-max	-	-	0.374	0.583	0.411	0.723
<i>Knowledge-guided models</i>						
PLATO	0.476	0.582	0.462	0.371	0.624	0.307
Ours	0.565	0.659	0.543	0.500	0.741	0.513

TabPFNv2 on most measures, achieving macro-AUPRC of 0.507 and macro-F1 of 0.488. Thus, the proposed framework offers its clearest advantage in AUPRC, macro-F1, and recall, rather than in every aspect of probability ranking or negative-case recognition. For the intended preliminary-screening task, the recall gain is central because higher macro-recall means that fewer positive disease labels are missed on average across the eight tasks. The macro-specificity of 0.513 shows the corresponding trade-off: the reduction in missed labels was accompanied by more false-positive predictions.

Direct LLM inference showed a higher-specificity, lower-recall operating point. Baichuan-M3 achieved the highest precision and specificity in the table, at 0.584 and 0.779, while GLM-5.2 obtained the best LLM macro-F1 of 0.418 and Qwen3.7-max the best LLM recall of 0.411. All three LLM recalls remained below 0.420. Relative to the best LLM recall (0.411) and macro-F1 (0.418), the proposed framework was higher by 0.330 and 0.125, respectively, but had lower precision and specificity. This pattern suggests more conservative positive decisions from direct LLM inference. Because the LLM comparison lacks continuous prediction scores, it supports an operating-point comparison only and does not establish a ranking-performance advantage over those models.

The low recall of direct LLM inference can be interpreted in light of the limited patient-level diagnostic evidence provided to the models. The reference labels were established from discharge diagnoses and examination reports, whereas direct LLM inference received only the structured screening variables and auricular measurements. For some diseases, key evidence used in clinical diagnosis was not included in the questionnaire or was represented only coarsely; missing questionnaire responses further reduced the patient-specific evidence available in some cases. Under this information asymmetry, an LLM applying general clinical knowledge may avoid assigning a positive label when explicit supporting findings are absent, which could contribute to more missed positive labels while preserving relatively high precision and specificity. Although the same input limitation applies to the supervised models, those models can learn cohort-specific proxy associations between the available variables and the labels, whereas the directly

Table 3. Component ablation results for multimodal knowledge enhancement. The best value for each metric is shown in bold.

Condition	Image	CDK	Macro-AUPRC	Macro-AUROC	Macro-F1	Macro-Precision	Macro-Recall	Macro-Specificity
M1: Common baseline	No	No	0.489	0.627	0.534	0.467	0.798	0.390
M2: Image only	Yes	No	0.504	0.612	0.538	0.464	0.816	0.367
M3: CDK only	No	Yes	0.509	0.626	0.545	0.470	0.855	0.348
M4: Full enhancement	Yes	Yes	0.513	0.624	0.547	0.466	0.883	0.315

prompted LLMs were not fitted to such local patterns. Taken together, this information gap is consistent with the high-specificity, low-recall operating point observed for direct LLM inference.

4.3 Ablation Studies

The ablation analysis first evaluates the two components that define multimodal knowledge enhancement and then examines the supporting training and decision components.

Component Ablation of Multimodal Knowledge Enhancement The four-condition ablation independently controls the ten auricular image-derived features and the clinical diagnostic knowledge topology. All conditions retain the questionnaire variables, background medical knowledge graph, PLATO architecture, class weighting, bidirectional sampling, DCT procedure, data partitions, and other training settings. The image-present conditions without and with CDK are identical to A4 and A7, respectively, in Table 4, so their evaluation results are reused rather than obtained from duplicate experiments. Table 3 reports the four combinations.

Without CDK, adding the auricular features increased macro-AUPRC by 0.015, macro-F1 by 0.004, and macro-recall by 0.018, while macro-AUROC and macro-specificity decreased by 0.015 and 0.023. With CDK present, the image-derived features produced smaller gains in macro-AUPRC and macro-F1 of 0.004 and 0.002, but increased macro-recall by 0.028; specificity decreased by 0.033. The image component therefore produced modest gains in macro-AUPRC, macro-F1, and macro-recall in both knowledge settings, with lower specificity.

CDK produced a larger recall effect. Without auricular features, adding CDK increased macro-AUPRC by 0.020, macro-F1 by 0.011, and macro-recall by 0.057, with a 0.042 decrease in specificity. When the image-derived features were retained, CDK increased macro-AUPRC by 0.009, macro-AUROC by 0.012, macro-F1 by 0.009, and macro-recall by 0.067, while specificity decreased by 0.052. Compared with the common baseline, full enhancement increased macro-AUPRC by 0.024, macro-F1 by 0.013, and macro-recall by 0.085. It achieved the highest macro-AUPRC, macro-F1, and macro-recall among the four conditions, but its specificity was 0.075 lower. The ablation pattern is consistent with recall-oriented contributions from both components, although the recall gain was accompanied by more false-positive predictions.

Training and Decision Component Ablation Because the MLP is already evaluated as a baseline method in Section 4.2, the controlled sequence in Table 4 focuses on seven

Table 4. Performance of the controlled ablation configurations. The best reported value in each metric is shown in bold.

ID	Components	Macro-AUPRC	Macro-AUROC	Macro-F1	Macro-Precision	Macro-Recall	Macro-Specificity
A1	None	0.482	0.587	0.308	0.269	0.403	0.607
A2	CW	0.513	0.613	0.494	0.466	0.710	0.514
A3	CW+BS	0.510	0.617	0.525	0.463	0.814	0.393
A4	CW+BS+DCT	0.504	0.612	0.538	0.464	0.816	0.367
A5	CDK	0.479	0.578	0.311	0.266	0.443	0.572
A6	CW+BS+CDK	0.517	0.623	0.531	0.464	0.852	0.357
A7	CW+BS+DCT+CDK	0.513	0.624	0.547	0.466	0.883	0.315

fixed-parameter PLATO configurations. All configurations use the background medical knowledge graph and the same input matrix containing the ten auricular image-derived variables. CW denotes class weighting, BS bidirectional hard-example sampling, DCT disease-specific classification thresholds, and CDK the clinical diagnostic knowledge topology. A1 is the unmodified PLATO backbone. A2 through A4 successively add CW, BS, and DCT. A5 adds CDK to A1, while A6 and A7 evaluate CDK together with the imbalance-aware components. Unlike the primary comparison, DCT is deliberately omitted from configurations whose component lists do not include it so that its contribution can be isolated. This sequence does not evaluate the contribution of the image-derived features.

Table 4 combines the component settings with the corresponding aggregate point estimates from the outer-fold evaluations and repeated initializations. The comparisons are descriptive and are not interpreted as significance tests.

Class weighting produced the largest observed single-step improvement. From A1 to A2, macro-recall increased from 0.403 to 0.710 and macro-F1 from 0.308 to 0.494; macro-AUPRC and macro-AUROC also rose by 0.031 and 0.026. Specificity decreased from 0.607 to 0.514. Adding bidirectional sampling in A3 further increased recall by 0.104 and F1 by 0.031, while the two ranking metrics changed by less than 0.005. These results indicate that the imbalance-aware components primarily improved positive-label detection, with a corresponding increase in false-positive predictions.

DCT changed the decision operating point rather than the underlying ranking. Comparing A3 with A4, macro-F1 increased by 0.013 and macro-recall by 0.002, whereas specificity decreased by 0.026. The small observed differences in AUPRC and AUROC should not be attributed to threshold selection because both metrics are calculated from continuous prediction scores. With CDK present, the A6-to-A7 comparison showed a larger threshold effect: macro-recall increased from 0.852 to 0.883 and macro-F1 from 0.531 to 0.547, while specificity decreased from 0.357 to 0.315.

The contribution of CDK depended on the training setting. Adding CDK to the unmodified backbone (A1 versus A5) changed macro-F1 by only 0.003 and macro-recall by 0.040, with slightly lower ranking metrics. Under class weighting and bidirectional sampling, however, A6 exceeded A3 by 0.007 in macro-AUPRC, 0.006 in macro-AUROC, 0.038 in macro-recall, and 0.006 in macro-F1. The complete preset configuration A7 achieved the highest macro-AUROC, macro-recall, and macro-F1 in the controlled sequence. Relative to A1, recall increased by 0.480 and F1 by 0.239, but specificity decreased by 0.292. Thus, the ablation pattern is consistent with a combined contri-

bution from imbalance-aware training and clinical diagnostic knowledge to reducing missed positive labels, while also showing the resulting recall-specificity trade-off.

5 Conclusion

This study developed a low-cost, non-invasive framework for preliminary multilabel screening of eight CHD-related conditions by combining a 52-item structured clinical assessment, ten auricular geometric variables, medical knowledge, imbalance-aware training, and disease-specific thresholds. In a retrospective cohort of 367 patients, the framework achieved the highest macro-AUPRC (0.565), macro-F1 (0.543), and macro-recall (0.741) among supervised comparators, whereas BR-TabPFNv2 achieved higher macro-AUROC, macro-precision, and macro-specificity. The principal observed benefit was therefore fewer missed positive labels rather than uniform superiority, with lower specificity and more false positives as the main trade-off. These findings provide preliminary evidence for the feasibility of low-cost knowledge-enhanced screening, but the retrospective single-center design, small cohort, severe label imbalance, few STEMI and NSTEMI cases, exploratory model selection, and inconsistent uncertainty estimates limit generalizability. Prospective multicenter validation with better representation of rare labels and independent determination of model settings and thresholds is required before clinical use.

Code Availability

The source code, model configurations, and data preprocessing scripts are publicly available at: <https://github.com/Laity163/MyPaper>.

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